

Dipraj Roy

Graduate Research Assistant | Geochemical Research Lab, University of Barishal, Bangladesh

Kornokathi, Barishal-8200, Bangladesh | +880 1792067975

diprajroy@gmail.com | diprajroy.com | ORCID: 0009-0003-4614-9345 | LinkedIn | ResearchGate | Google Scholar | Web of Science

Profile

Early-career geoscientist specializing in sedimentary and environmental geochemistry, coastal hazard vulnerability, and GIS/remote-sensing-based geospatial analysis. Co-author of six manuscripts within two years of research — three published in Elsevier Q1 journals, one under review at Elsevier, and two recently submitted (including a first-authored study at *City and Environment Interactions* and a co-authored study at ACS's *Environmental Science & Technology*) — with an active pipeline spanning provenance geochemistry, machine-learning-based hydrogeomorphology, and multi-criteria vulnerability assessment. National Science and Technology (NST) Fellow 2026, currently seeking PhD opportunities for higher studies abroad.

Education

Master of Science, Geology and Mining

University of Barishal, Bangladesh

CGPA: 3.70/4.00

Jan 2025 – Apr 2026

Thesis: Geochemical Composition of the Brahmaputra River Sediments: Implications for Provenance, Weathering, Tectonic Setting, Climatic Influence, Depositional Environment and REEs Potential

Bachelor of Science, Geology and Mining

University of Barishal, Bangladesh

CGPA: 3.62/4.00

Jan 2019 – Dec 2024

Higher Secondary School Certificate (Science)

Govt. Bangabandhu College, Rupsha, Khulna

CGPA: 5.00/5.00

Result: Jul 2018

Secondary School Certificate (Science)

Betaga United Secondary School, Fakirhat, Bagerhat

CGPA: 5.00/5.00

Result: May 2016

Research Interests

Sedimentary and environmental geochemistry | Provenance studies | Heavy metal and radioactivity assessment | GIS and remote sensing | Coastal and vulnerability hazard analysis | Atmospheric and climate change studies | Carbon sequestration | Geohazards and anthropogenic hazards

Publications

- [Submitted] Roy, D., Paul, S., Guha, B., Halder, M., Aishe, D.S., Pompa, S.C., Khan, A., Khan, R., Roy, D.K.* (2026). *Spatially Explicit Waterlogging Risk Assessment for Climate-Resilient Urban Planning: A RS-GIS-AHP Framework for Two Major Coastal Cities in Bangladesh*. *City and Environment Interactions* (Manuscript ID: CACINT-D-26-00683).
- [Submitted] Aishe, D.S., Khan, R., Roy, D., Pranto, A.I., Kundu, Z., Alam, M.S., Hossain, F., Islam, M.S., Tamim, U., Mallick, J., Hang, H.T., Roy, D.K. (2026). *Integrated Elemental Geochemistry Reveals Natural and Anthropogenic Controls on Himalayan-Derived Sediments in the Transboundary Punarbhaba River*. *Environmental Science & Technology* (ACS, Manuscript ID: es-2026-10465g).
- [Under Review] Roy, D., Khan, R., Aishe, D.S., Kundu, Z., Pranto, A.I., Alam, M.S., Tamim, U., Dey, A., Nafiz, J., Mallick, J., Hang, H.T., Roy, D.K.* (2026). *Geochemical fingerprints of Himalayan sediment dispersal in the Brahmaputra River system: provenance, weathering, tectonic setting, climate and rare earth element enrichment*. *Physics and Chemistry of the Earth* (Elsevier, Q1).
- Kundu, Z., Khan, R., Roy, D., Pranto, A.I., Aishe, D.S., Ibrahim, K.A., Islam, M.S., Idris, A.M., Naher, K., Roy, D.K. (2026). *Studying the geochemical processes of environmental radionuclide alterations in a coastal area in terms of lanthanides: Probabilistic radiological risk assessment*. *Marine Pollution Bulletin*, 230, 119896. doi:10.1016/j.marpolbul.2026.119896 (Elsevier, Q1).

- Aktar, S., Hossain, M., Rakib, M.M.H., Saha, S.C., Ovi, M.H., **Roy, D.**, Khan, A., Hasan, I., Salman, M.A., Khan, R., Rafiq, M.R.*, Roy, D.K.** (2025). *Geomorphological responses to climate change in the Ganges-Brahmaputra-Meghna Delta: A multi-decadal remote sensing analysis*. Quaternary Science Advances, 100, 100296. doi:10.1016/j.qsa.2025.100296 (Elsevier, Q1).
- Roy, D.K.*, Khan, R., Kundu, Z., Ornee, T.I., Rakib, M.M.H., Ovi, M.H., **Roy, D.**, Pranto, A.I., Aishe, D.S., Abdullah, M., Nandi, N., Islam, F., Khan, M.H.R., Aldawood, S., Idris, A.M., Khan, A., Saha, A., Alam, M.S. (2025). *Major, trace, and rare earth element geochemistry of the Kuakata beach sand, Bangladesh: Implications for provenance, weathering, tectonic setting, and heavy metal contamination*. Regional Studies in Marine Science, 89, 104338. doi:10.1016/j.rsma.2025.104338 (Elsevier, Q1).

Conference Paper

- Halder, M., Guha, B., Paul, S., **Roy, D.*** (2025). *AHP-based Multi-criteria Assessment of Drought Vulnerability in Rangpur Division, Bangladesh*. International Conference on Environment & Changing Climate 2025, Independent University, Bangladesh (Oral Presentation).

Research Experience

Waterlogging Risk Analysis of Two Coastal Cities of Bangladesh Jul 2025 – Present
 Dept. of Geology and Mining, University of Barishal

- Used 12 key parameters to identify waterlogging risk with AHP, validated using ROC-AUC; findings submitted as first author to City and Environment Interactions (Manuscript ID: CACINT-D-26-00683).

Geochemical Composition of the Brahmaputra River Sediments Jan 2025 – May 2026
 Dept. of Geology and Mining, University of Barishal

- Led geochemical characterization of river sediments to reconstruct provenance, weathering intensity, tectonic setting, and paleoclimatic/depositional controls using major, trace, and REE proxies.

Multi-criteria Analysis of Cyclone Vulnerability, Cox's Bazar Dec 2024 – Present
 Dept. of Geology and Mining, University of Barishal

- Applying a geospatial AHP-based multi-criteria framework to map cyclone vulnerability along the Cox's Bazar coastline.

Kuakata Beach Sand Geochemistry Oct 2024 – Jul 2025
 Dept. of Geology and Mining, University of Barishal

- Investigated major, trace, and rare earth element geochemistry to characterize provenance, weathering, tectonic setting, and heavy-metal contamination; findings published in Regional Studies in Marine Science.

GBM Delta Geomorphological Response to Climate Change Oct 2024 – Aug 2025
 Dept. of Geology and Mining, University of Barishal

- Conducted a multi-decadal remote sensing analysis of geomorphological change across the Ganges-Brahmaputra-Meghna Delta; findings published in Quaternary Science Advances.

Field Research Experience

- **Sample Collection (Jan – Mar 2025):** Collaborated in a multidisciplinary field team across three river systems (Lower Meghna, Kirtonkhola, Sandha-Katcha-Baleshwar), managing site navigation, sample collection, and documentation for geochemical, heavy-metal, and radiological assessments.
- **Geological Mapping & Sedimentological Assessment (Yr 1–4 undergraduate field work):** Conducted outcrop inspection, fossil/mineral/rock identification, structural investigation, geological mapping, and data interpretation across the Eastern Hill Tracts and coastal Bengal Basin.

Professional & Other Experience

Intern, Research & Development Unit Since Jul 2026
 UniV Bangladesh

Trainee Assistant Officer (Cash) Since Jun 2026
 Southeast Bank PLC, Bangladesh

Graduate Research Assistant Since Nov 2024
 Geochemical Research Lab, University of Barishal

Awards & Achievements

- National Science and Technology (NST) Fellowship, 2026.
- **Best Paper Award**, International Conference on Environment & Changing Climate 2025 (IUB, in partnership with COLOCAL Project) — “AHP-based Multi-criteria Assessment of Drought Vulnerability in Rangpur Division, Bangladesh.”
- **Finalist**, 3rd International Poster Competition for Young Researchers 2025 — “Assessment of Waterlogging Vulnerability in Coastal Cities of Bangladesh: A Comparative AHP-Based Study of Khulna and Barishal” (Organized by UNIV).
- **Finalist**, Water Hackathon 2025 (BRAC & IRC), for an innovative water and sanitation solution (Jul 2025).
- Published article, “COVID-19: Blessing for Environment,” in *Silent Smile of the Green World: Covid 19 and Our Future* (ISBN 978-984-35-1407-3).

Certifications

- Computer Fundamentals and Office Applications (Word, PowerPoint, Excel) — Enhancing Digital Government & Economy Project (Aug–Oct 2024, 60 hrs).
- Automate the Boring Stuff with Python Programming — Udemy, taught by Al Sweigart (2021).
- Excel Skills for Business: Essentials — Macquarie University, via Coursera.
- Oil & Gas Industry Operations and Markets — Duke University, via Coursera.

Conferences & Seminars

- Conference on Environment & Changing Climate 2025: Global Strategies, Local Actions (IUB & COLOCAL, Nov 2025) — presented “AHP-based Multi-criteria Assessment of Drought Vulnerability in Rangpur Division, Bangladesh.”
- Symposium on Innovative Energy Infrastructure: Shaping a Clean and Affordable Future for Bangladesh (Dept. of Geology and Mining, BU, with BEPRC, Sep 2025).
- Seminar: “Geo-Information for Climate Change Resilient Urban Planning and Development of Barishal City” (Dept. of Geology and Mining, BU, GSB, and BGR Germany, Apr 2025).

Technical Skills

- **Programming:** Python
- **Geospatial Tools:** ArcGIS Pro, ArcGIS 10.8, Google Earth Engine
- **Modelling & Plotting:** Origin Pro, KaleidaGraph, EPA PMF 5.0
- **Office Tools:** Word, Excel, PowerPoint
- **Web & App Development:** Flutter, HTML, WordPress

References

Dr. Dhiman Kumer Roy
Associate Professor & Chairman
Dept. of Geology and Mining
University of Barishal
dhimans14@yahoo.com

Dr. Atikul Haque Farazi
Assistant Professor
Dept. of Geology and Mining
University of Barishal
ahfarazi@bu.ac.bd

Sukhen Goswami
Associate Professor
Dept. of Geology and Mining
University of Barishal
swami.sg.du@gmail.com



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